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Slide material is largely sourced from Slack & Bandon-Jones (2019) *Operations Management.* Ninth Edition (Pearson), and Heizer, Render & Munson (2019) *Operations Management*. Twelfth Edition (Pearson)

Process Strategy

Lecture 3

Dr Roberto Chavez

SWINBURNE UNIVERSITY OF TECHNOLOGY

Acknowledgement of Country

We respectfully acknowledge the Wurundjeri People of the Kulin Nation, who are the Traditional Owners of the land on which Swinburne's Australian campuses are located in Melbourne's east and outer-east, and pay our respect to their Elders past, present and emerging.

We are honoured to recognise our connection to Wurundjeri Country, history, culture, and spirituality through these locations, and strive to ensure that we operate in a manner that respects and honours the Elders and Ancestors of these lands.

We also respectfully acknowledge Swinburne's Aboriginal and Torres Strait Islander staff, students, alumni, partners and visitors.

We also acknowledge and respect the Traditional Owners of lands across Australia, their Elders, Ancestors, cultures, and heritage, and recognise the continuing sovereignties of all Aboriginal and Torres Strait Islander Nations.

OUTLINE

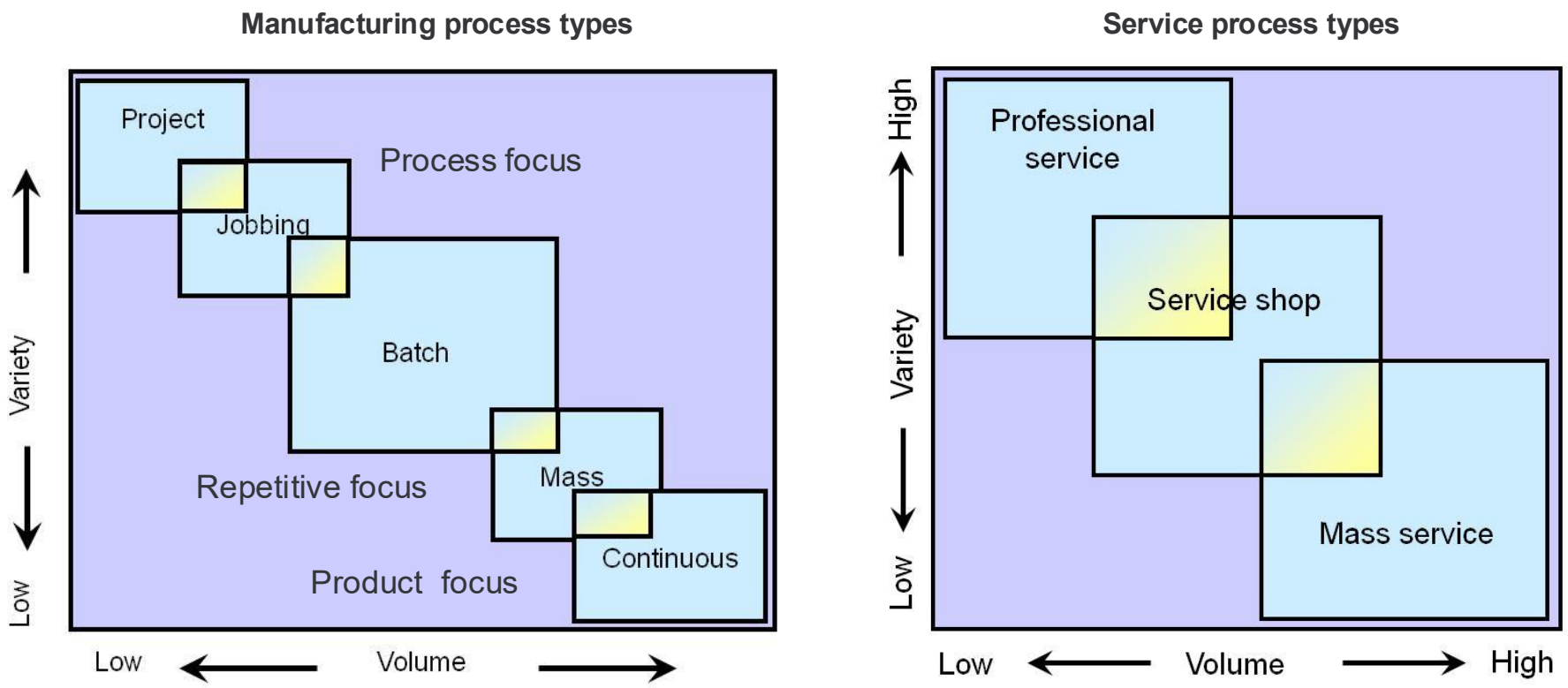
- Process strategies
- The strategic importance of layout decisions
- Types of Layout

PROCESS STRATEGIES

Four general strategies/types:

1. Process focus
2. Repetitive focus
3. Product focus
4. Mass customization

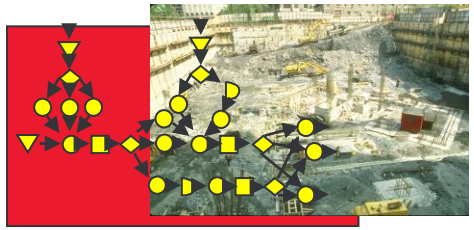
THE VOLUME-VARIETY EFFECT



Source: Slack et al. (2007)

PROCESS FOCUS: PROJECT, JOBBING

- Facilities are organized around specific activities/projects or processes (products)
- General purpose equipment and skilled personnel
- High degree of product flexibility
- Typically high costs and low equipment utilization
- Product flows may vary considerably making planning and scheduling a challenge

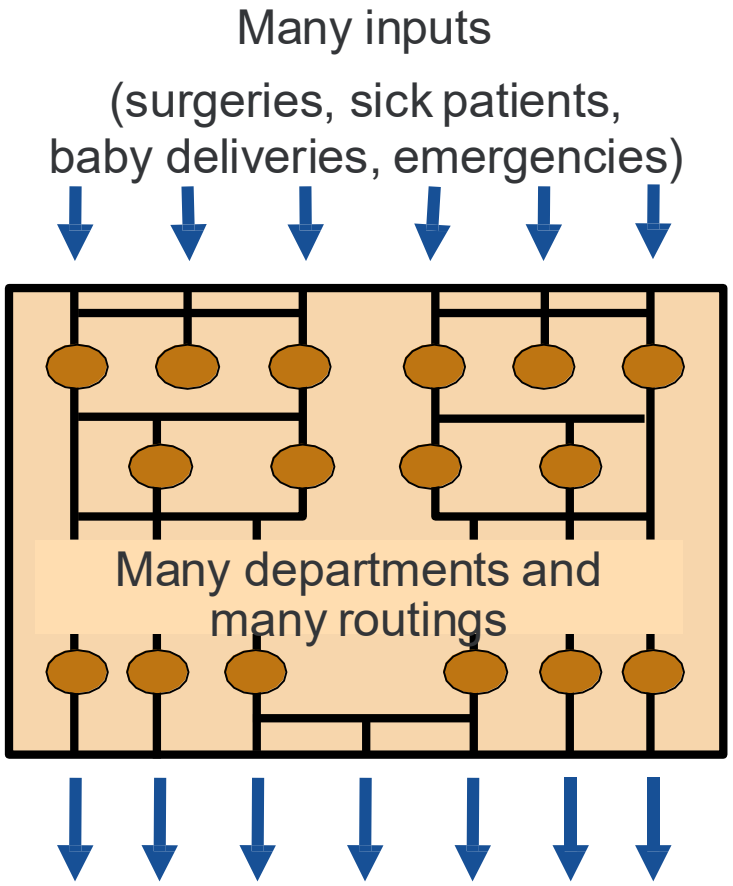


Source: Slack et al. (2007)

PROCESS FOCUS

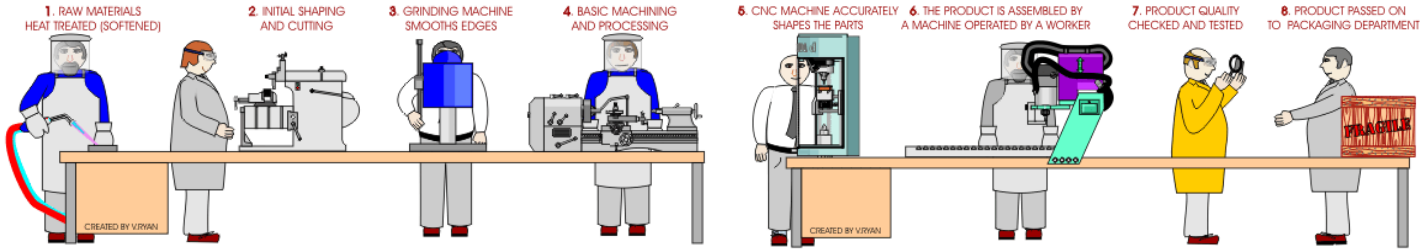
(low-volume, high-variety, intermittent processes)

e.g. Hospital



REPETITIVE FOCUS : BATCH, MASS

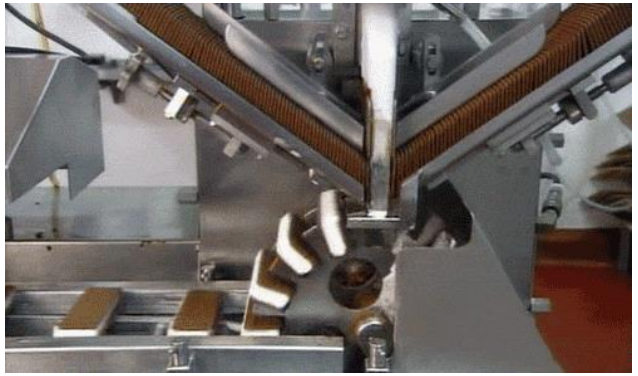
- Facilities are often organized as assembly lines
- Characterized by modules with parts and assemblies made previously
- Modules may be combined for many output options
- Less flexibility than process-focused facilities but more efficient use of resources



Source: Ryan, V. Batch Production. Digital image. Batch Production. N.p., n.d. Web. 21 Jan. 2013.

PRODUCT FOCUS: MASS, CONTINUOUS

- High volume but low product variety
- Long, continuous production runs enable efficient processing
- Typically, high fixed cost (high automation) and low variable cost
- Generally, less skilled labour (specialization)



Source: Slack et al. (2007)

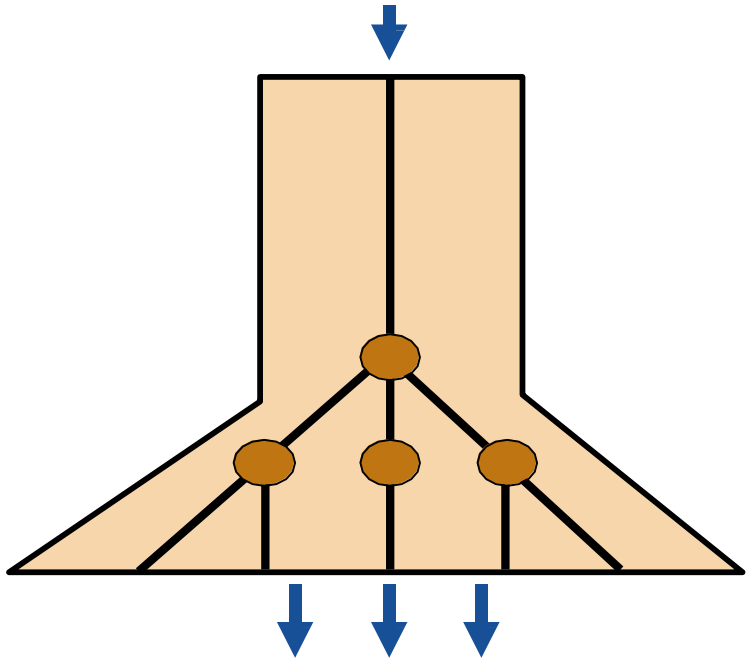
PRODUCT FOCUS

(high-volume, low-variety, continuous process)

e.g., Frito-Lay

Few inputs

(corn, potatoes, water, seasoning)



Output variations in size, shape, and packaging

(Package labeled for each material)

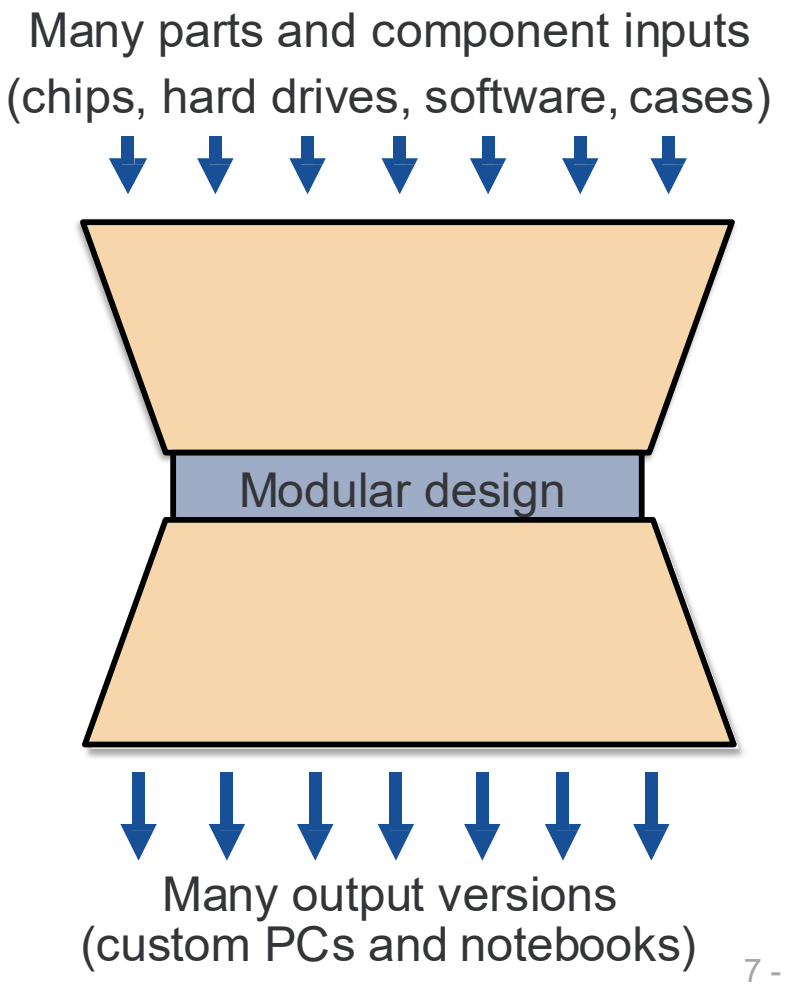
MASS CUSTOMIZATION

- The rapid, low-cost production of goods and service to satisfy increasingly unique customer desires
- Combines the flexibility of a process focus with the efficiency of a product focus
- However, this requires standardised design: process, products, or components and customization capabilities.

MASS CUSTOMIZATION

(high-volume, high-variety)

e.g., Dell



MASS CUSTOMIZATION

TABLE 7.1	Mass Customization Provides More Choices Than Ever	
	NUMBER OF CHOICES	
ITEM	1970s	21ST CENTURY
Vehicle styles	18	1,212
Bicycle types	8	211,000
iPhone mobile game apps	0	1,200,000
Web sites	0	634,000,000
Movie releases per year	267	1551
New book titles	40,530	300,000+
Houston TV channels	5	185
Breakfast cereals	160	340
Items (SKUs) in supermarkets	14,000	150,000
High-definition TVs	0	102

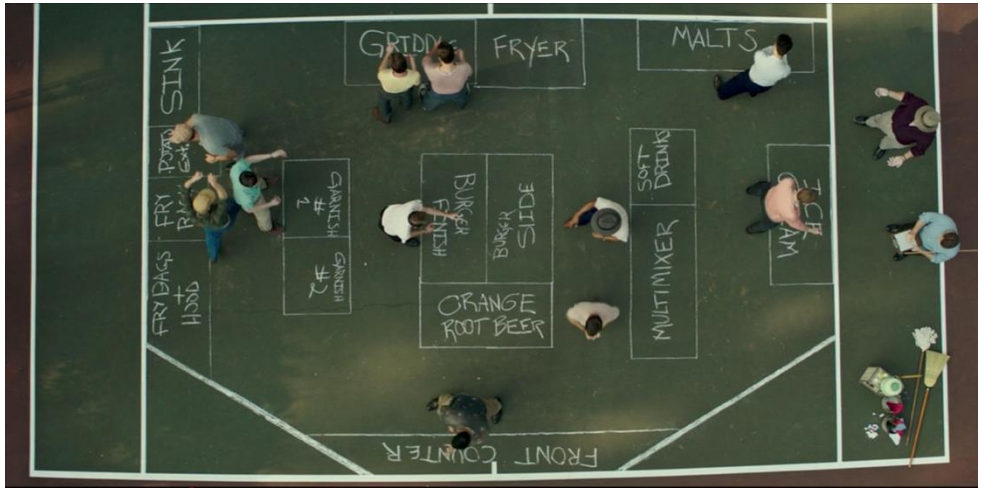
PROCESS STRATEGIES: CHARACTERISTICS

TABLE 7.2 Comparison of the Characteristics of Four Types of Processes

PROCESS FOCUS (LOW-VOLUME, HIGH-VARIETY ARNOLD PALMER HOSPITAL)	REPETITIVE FOCUS (MODULAR HARLEY-DAVIDSON)	PRODUCT FOCUS (HIGH-VOLUME, LOW-VARIETY FRITO-LAY)	MASS CUSTOMIZATION (HIGH-VOLUME, HIGH-VARIETY DELL COMPUTER)
1. Small quantity and large variety of products	1. Long runs, a standardized product from modules	1. Large quantity and small variety of products	1. Large quantity and large variety of products
2. Broadly skilled operators	2. Moderately trained employees	2. Less broadly skilled operators	2. Flexible operators
3. Instructions for each job	3. Few changes in the instructions	3. Standardized job instructions	3. Custom orders requiring many job instructions
4. High inventory	4. Low inventory	4. Low inventory	4. Low inventory relative to the value of the product

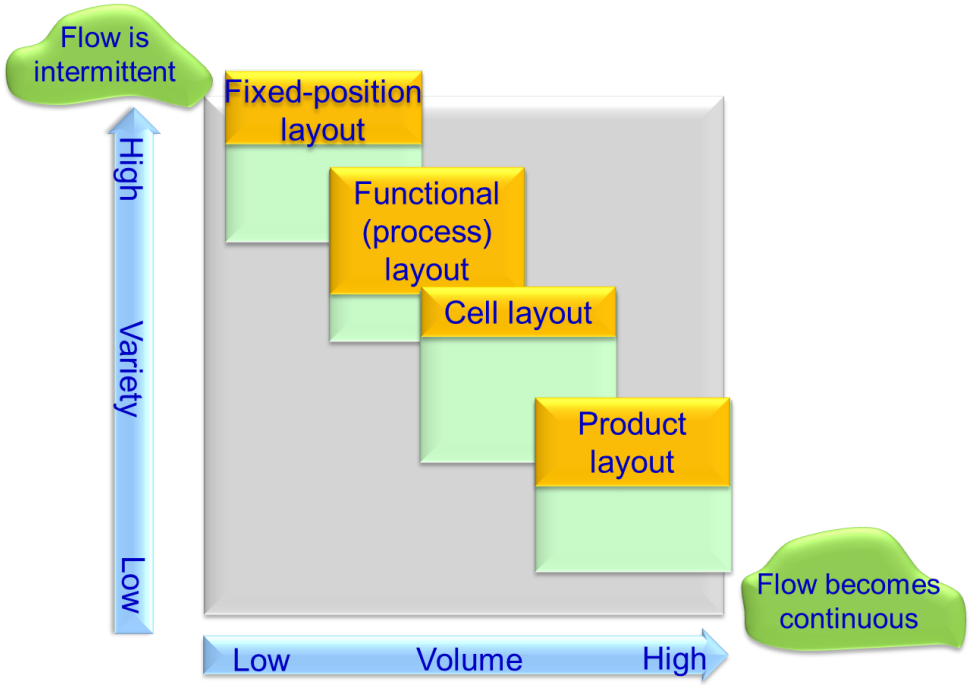
LAYOUT AND FLOW

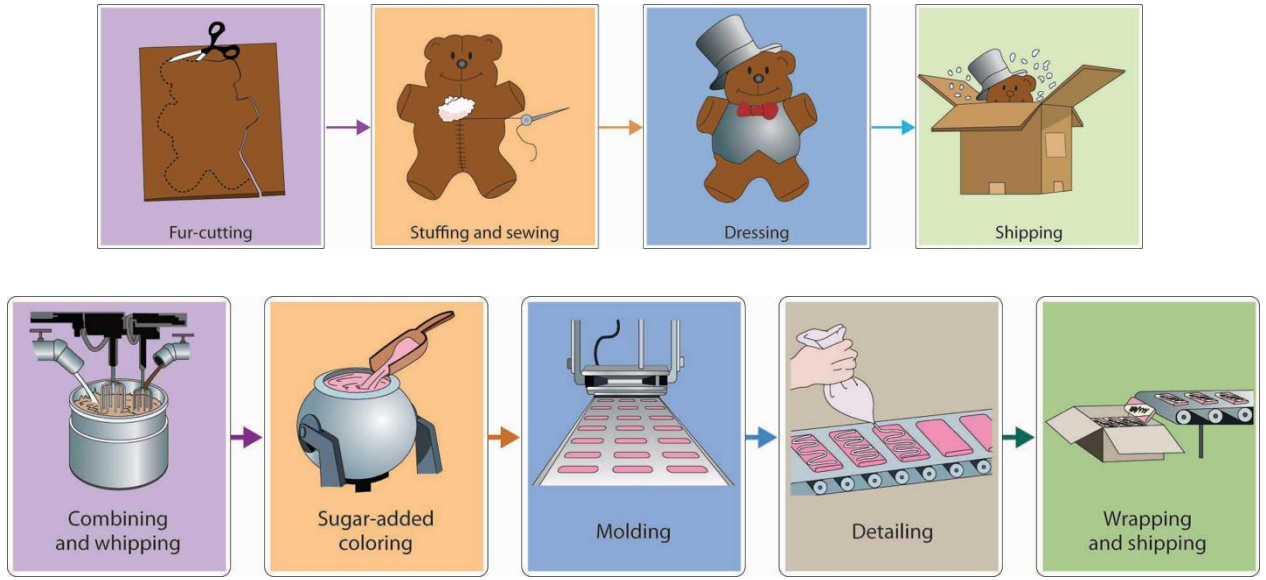
- Layout of an operations is concerned with the physical representation of a process
- It considers resource utilization:
  - Machines
  - Equipment
  - Staff
  - Transformed resources
- The objective is to maximise the operation/use of resources (e.g. reduce movement, cost and increase effectiveness)
  - Creating a process that meets customer requirements within cost and other managerial constraints (time, cost, energy etc.)



Source: https://www.subtraction.com/2018/01/23/design-lessons-from-mcdonalds/

LAYOUT TYPE: VOLUME & VARIETY





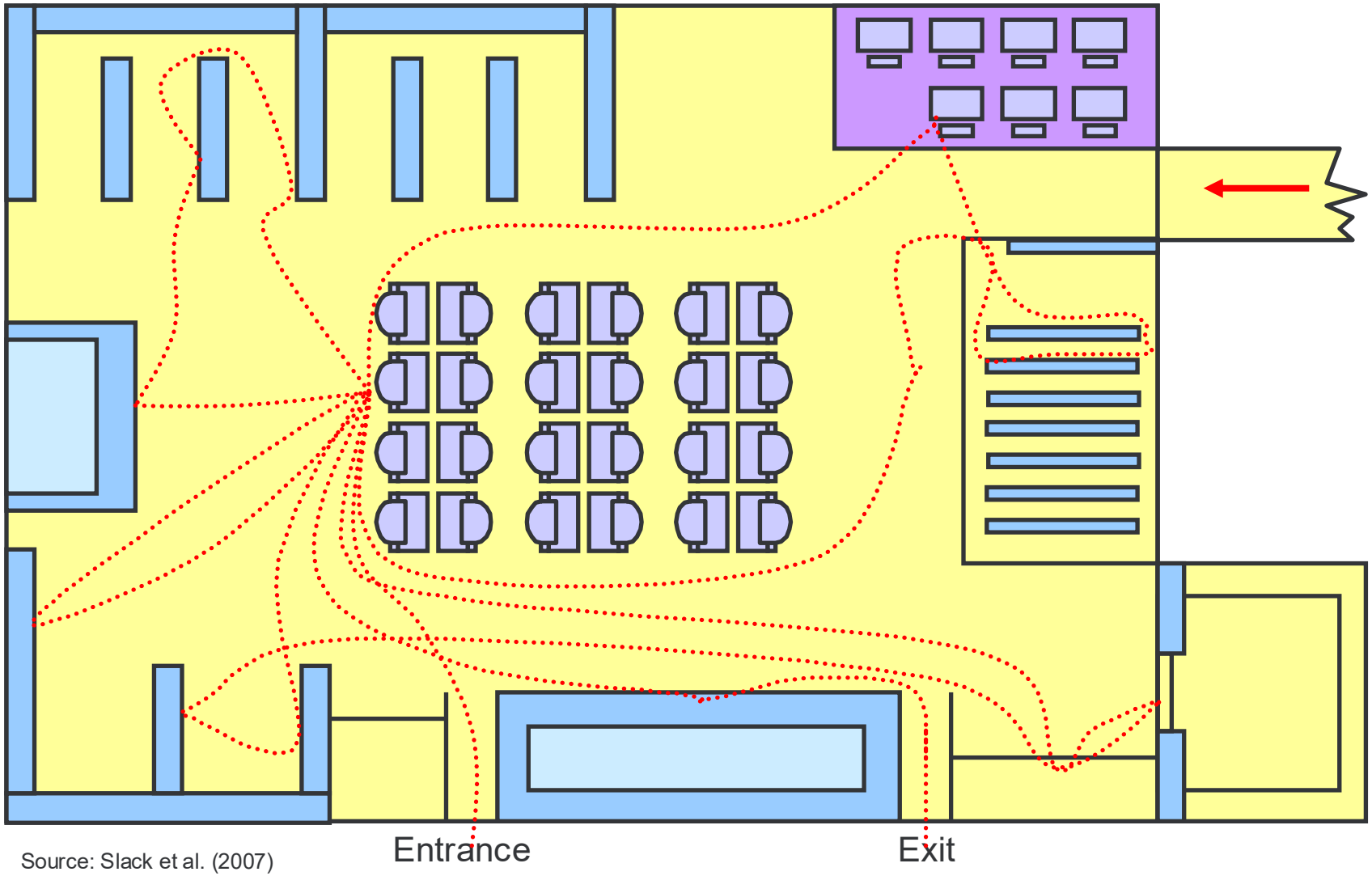
Source:https://open.lib.umn.edu/exploringbusiness/chapter/11-2-facility-layouts/

LAYOUT CHARACTERISTICS

Layout type	Characteristics
Fixed-position layout	The transforming resources move where the transformation process is taking place (e.g. Dams)
Functional (Process) layout	Products move freely according to its needs (e.g. Patients in an emergency room)
Cell layout	Products move freely according to its needs but there is a predetermined route (e.g. shopping malls)
Product layout	Product moves through the transforming resources following a predetermined route (e.g. manufacturing of electronic components)

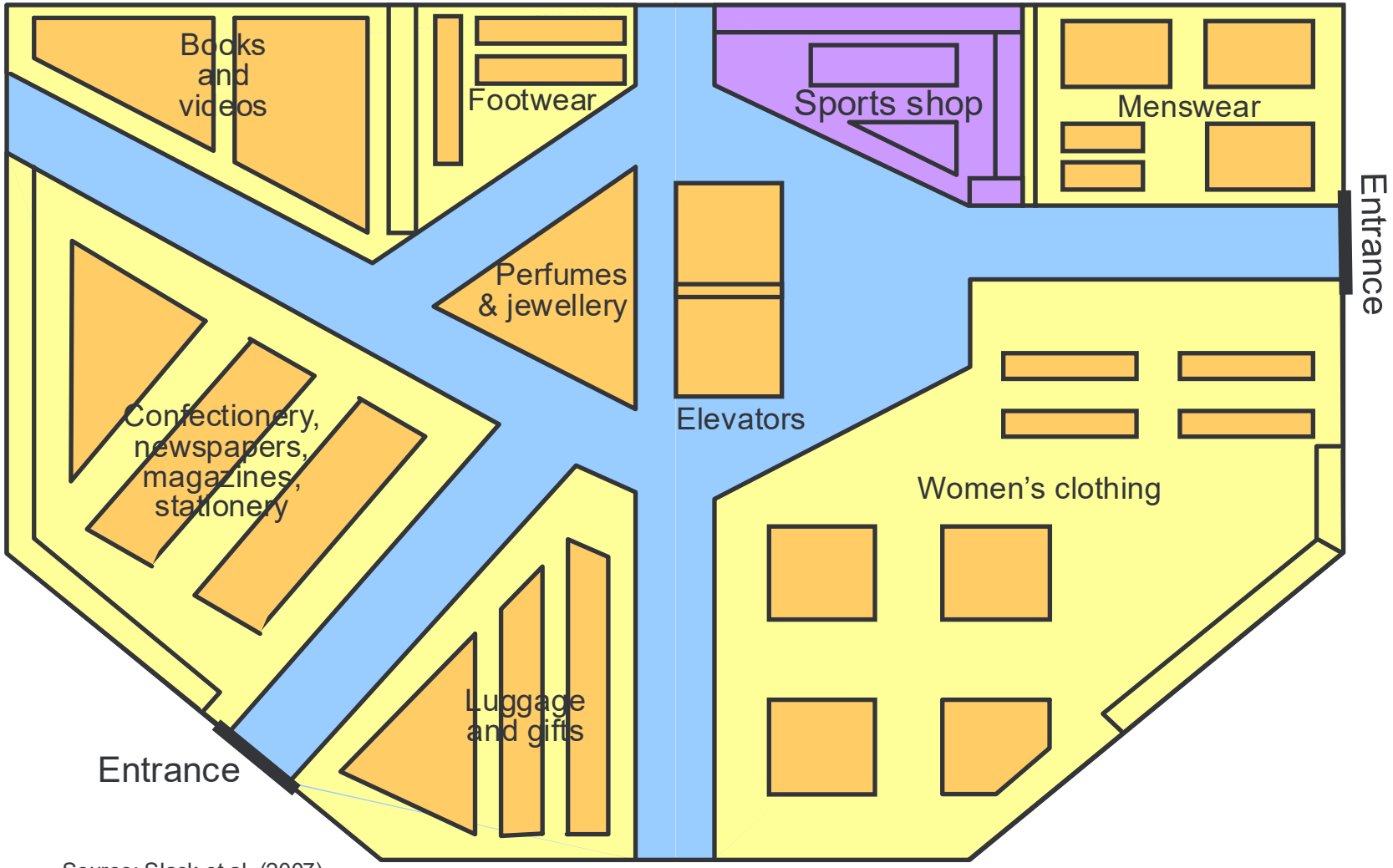
Source: Slack et al. (2007)

EXAMPLE OF A LAYOUT: LIBRARY



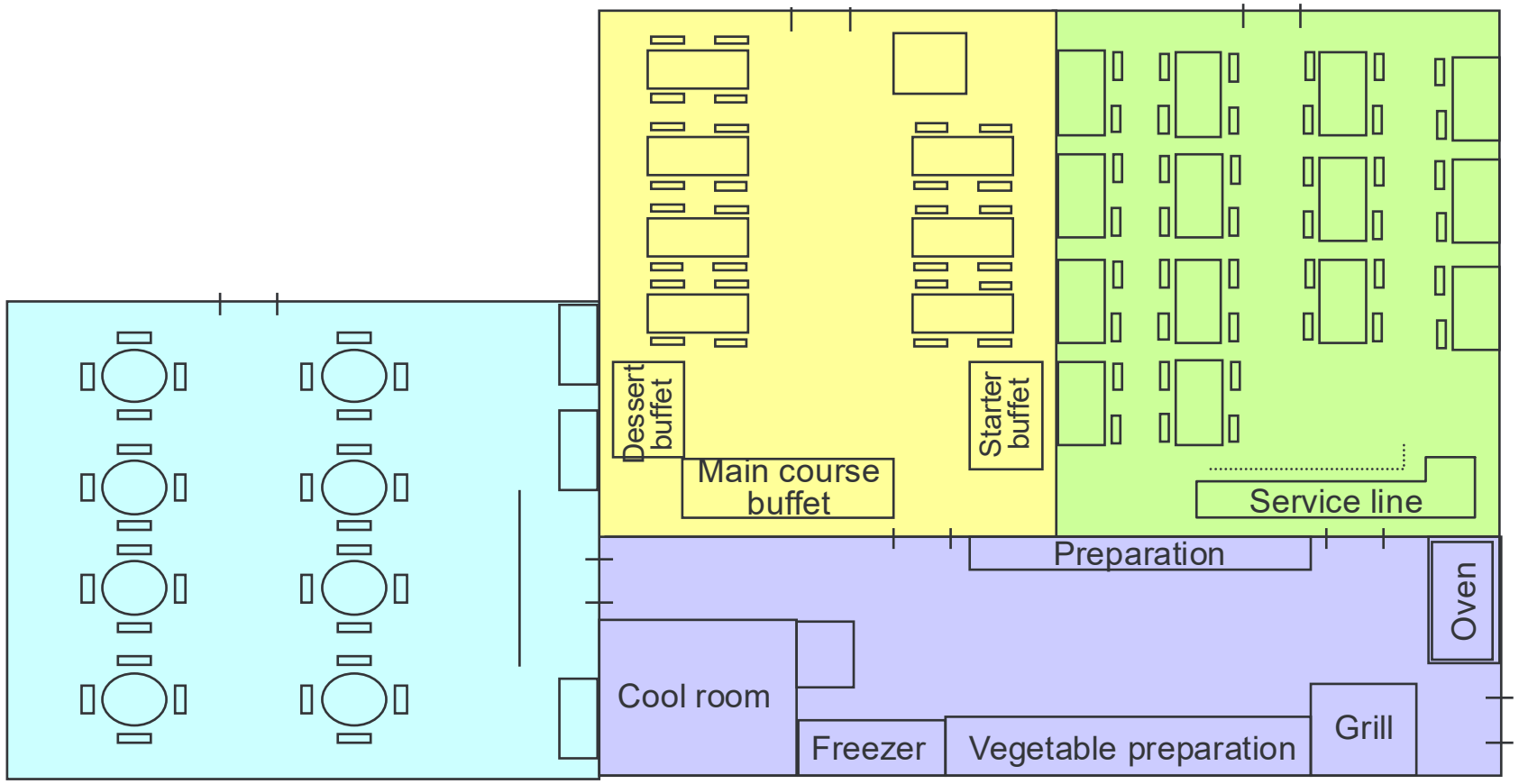
Source: Slack et al. (2007)

EXAMPLE OF A LAYOUT: DEPARTMENT STORE



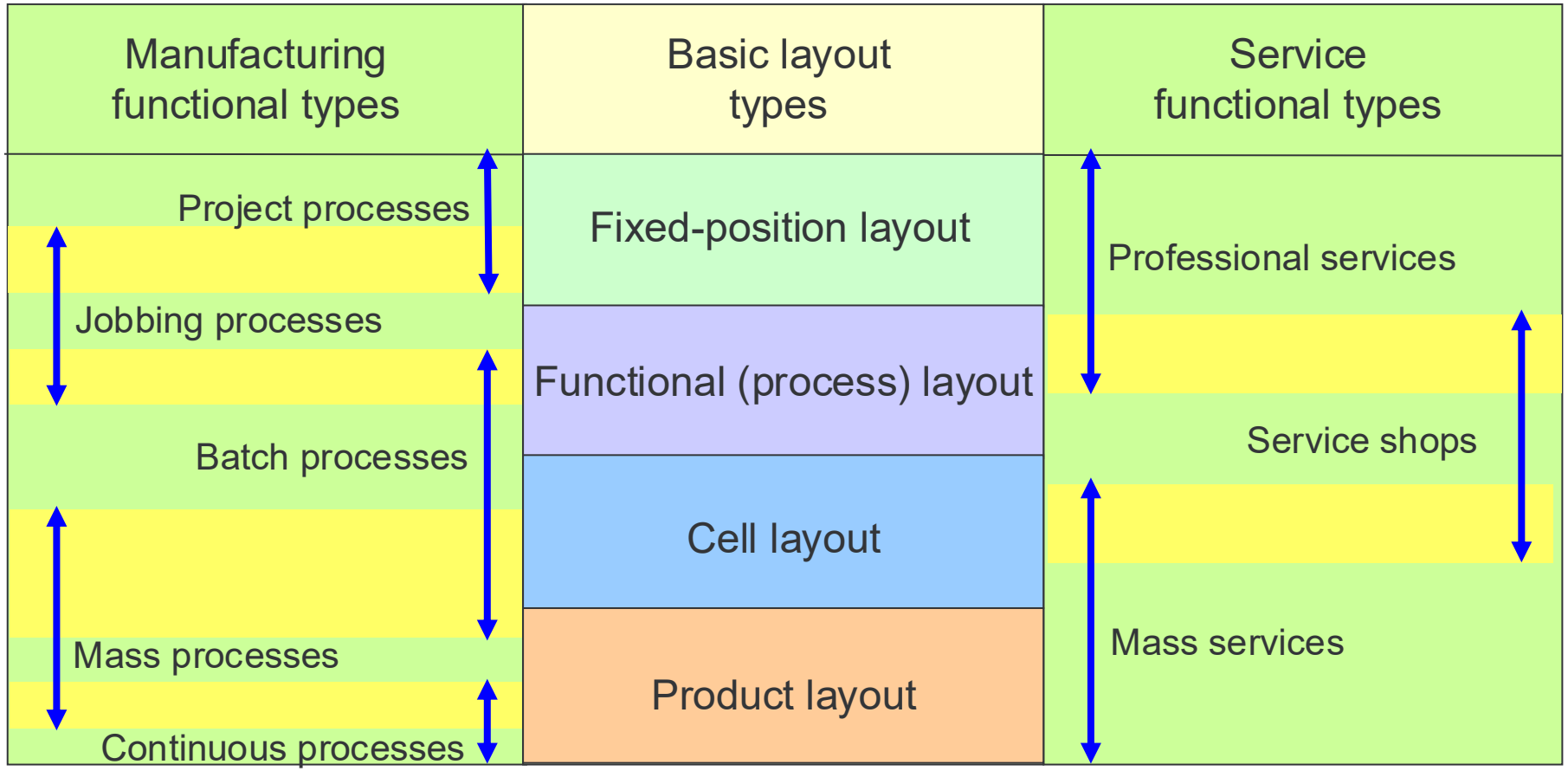
Source: Slack et al. (2007)

A RESTAURANT COMPLEX WITH ALL FOUR BASIC LAYOUT TYPES



LAYOUT AND PROCESS

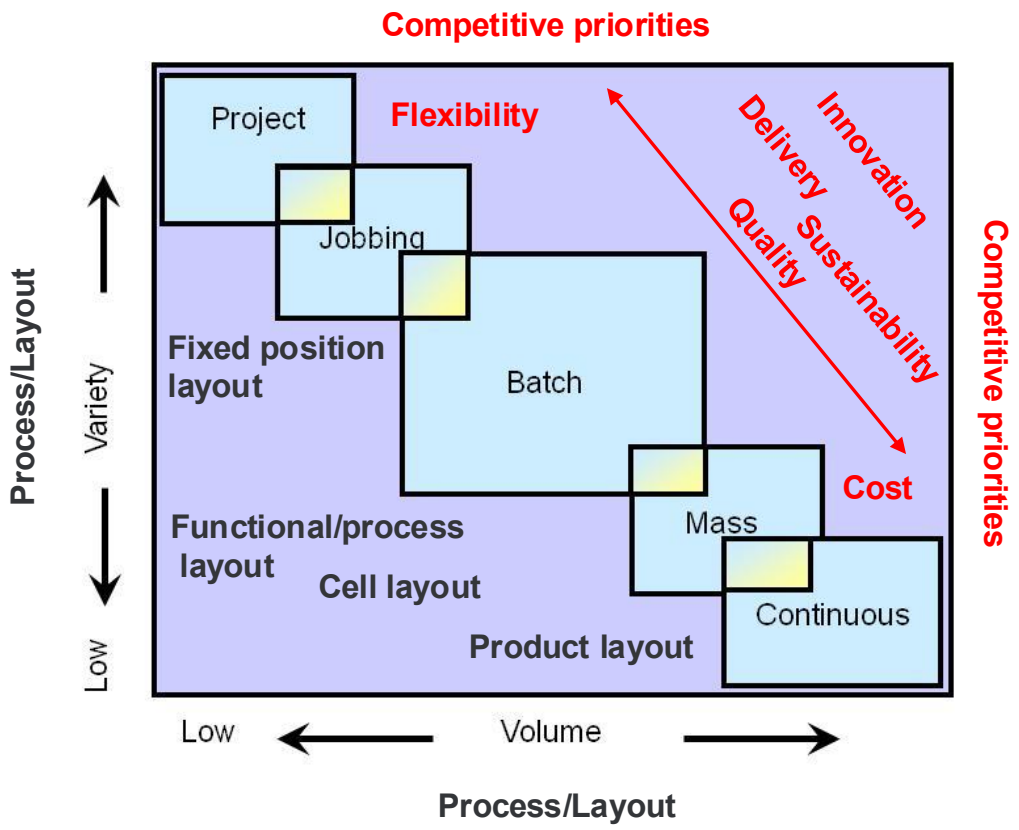
- Layouts are clearly linked to processes



Source: Slack et al. (2007)

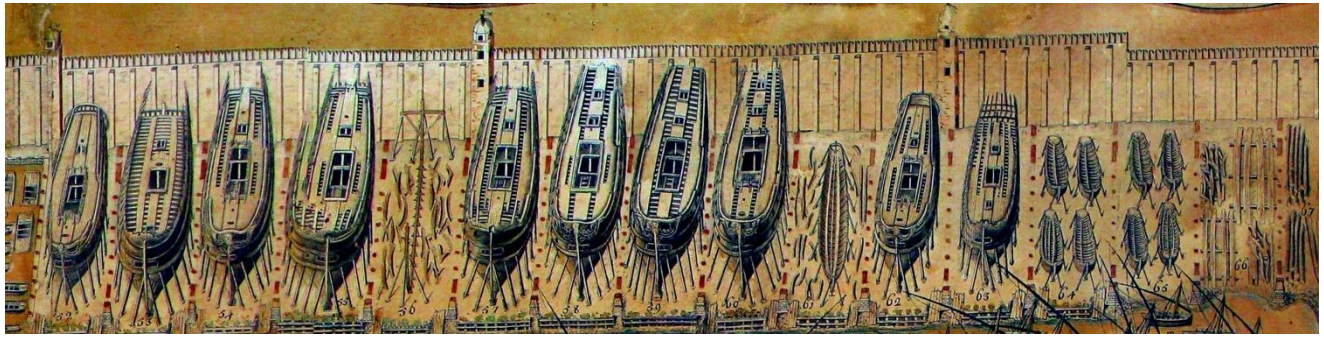
PROCESS/LAYOUT & COMPETITIVE PRIORITIES

- Pick a specific product or service that you are familiar with and explain what would be a suitable process type and layout type.



THEORY OF SWIFT EVEN FLOW (TSEF)

TSEF proposes that "…the more swift and even the flow of materials (or information) through a process, the more productive the process…" (Schmenner, 2004, p. 335) and conversely, "… [productivity] falls with increases in the variability associated with the flow…" (Schmenner, 2004, p. 335)



Source: Material Flow in the Arsenal of Venice 1797 | AllAboutLean.com

Source: The Cotton Industry - Historic UK

DIGITAL TRANSFORMATION AND PROCESS MANAGEMENT

Firms are just beginning to realize the potential of combining process management with digital technologies (AI, Biga Data, IoT, etc.)

Discuss in groups the following questions:

- What are the benefits of Process management and Digital technologies working together?
- What are the challenges?
- Bring one example from the industry to illustrate your ideas.

Source: https://www.isems.com.sg/digital-project-management

REFERENCES

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- Operations Management: Sustainability and Supply Chain Management, 2017, 12th Edition By Heizer, J. Render, B., Mundson, C.
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